MA 17: HOW TO SOLVE IT
HANDOUT 3: FUNCTIONS AND ELEMENTARY ANALYSIS

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Base representations: from last time. For any integer $b \geq 2$, every positive integer has a unique representation

$$n = a_r b^r + a_{r-1} b^{r-1} + \cdots + a_1 b + a_0, \qquad a_i \in \{0, 1, \ldots, b-1\}, \quad a_r \neq 0.$$

The cases $b = 2$ and $b = 10$ give binary and decimal representations.

Problem 1. (Handout 2, Problem 8)
Let $k \geq 2$ be an integer. Find the sum of all integers n with $1 \leq n \leq 2^k - 1$ whose binary representation contains an even number of ones.
Hint: Start with $k = 2, 3$. Include 0 and group consecutive integers: $(0, 1, 2, 3), (4, 5, 6, 7) \ldots$.

Problem 2.
Prove that for every positive integer n, there exists an n-digit positive integer (in decimal representation) divisible by 2^n, such that each of its decimal digits is either 2 or 3.
Hint: There are 2^n candidates. Show that precisely one of them works.

Continuity. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Continuity at $a \in D$ means that $f(x_n) \to f(a)$ for every sequence (x_n) in D with $x_n \to a$. The function f is continuous on D if it is continuous at every point of D.

Problem 3.
Find every continuous function $g : \mathbb{R} \to \mathbb{R}$ satisfying $g(0) = 1$ and

$$g(2x) - g(x) = x \qquad \text{for every real } x.$$

Hint: Try a function of the form $g(x) = cx + d$. To prove uniqueness, subtract your candidate from an arbitrary solution.

The intermediate value theorem. If f is continuous on $[u, v]$, where $u < v$, then it takes every value between $f(u)$ and $f(v)$.

Problem 4.
Let $f : [0, 1] \to \mathbb{R}$ be continuous, with $f(0) = f(1)$. Prove that for every positive integer n, there is an $x \in [0, 1 - 1/n]$ such that

$$f(x) = f\left(x + \frac{1}{n}\right).$$

Hint: Consider the values of f at $0, 1/n, 2/n, \ldots, 1$. Apply the intermediate value theorem to a suitable difference function.

Derivatives. For a function defined on an open interval, its derivative at x is

$$\frac{d}{dx} f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

when this limit exists and is finite. Differentiability implies continuity. For differentiable functions, the product and chain rules state that

$$(fg)' = f'g + fg', \qquad \frac{d}{dx}f(g(x)) = f'(g(x))g'(x).$$

A differentiable function with non-negative (resp. non-positive) derivative throughout an interval is non-decreasing (resp. non-increasing) there.

Problem 5. (Putnam 2023, A1)
For a positive integer n, let $f_n(x) = \cos(x)\cos(2x)\cos(3x)\cdots\cos(nx)$. Find the smallest n such that $|f_n''(0)| > 2023$.
Hint: Write $f_n(x) = f_{n-1}(x)\cos(nx)$ for $n \geq 2$.

Problem 6. (Putnam 2021, A2)
For every positive real number x, let

$$g(x) = \lim_{r\to 0}((x+1)^{r+1} - x^{r+1})^{\frac{1}{r}}.$$

Find $\lim_{x\to\infty}\frac{g(x)}{x}$.
Hint: Keep x fixed and set $A(r) = (x+1)^{r+1} - x^{r+1}$. Since $A(0) = 1$, what derivative is represented by $\lim_{r\to 0}\ln A(r)/r$?

Problem 7. (Putnam 1997, B2)
Let f be a twice-differentiable real-valued function satisfying

$$f(x) + f''(x) = -xg(x)f'(x),$$

where $g(x) \geq 0$ for all real x. Prove that $|f(x)|$ is bounded.
Hint: Multiply the equation by $f'(x)$. Can the left-hand side be written as a derivative? Consider the two sides of 0 separately.

Rolle's theorem. Suppose $u < v$ and H is continuous on $[u, v]$ and differentiable on (u, v). If $H(u) = H(v)$, then $H'(c) = 0$ for some $c \in (u, v)$.

Problem 8. (Putnam 1992, A4)
Let f be an infinitely differentiable real-valued function defined on the real numbers. If

$$f\left(\frac{1}{n}\right) = \frac{n^2}{n^2+1}, \qquad n = 1, 2, 3, \ldots,$$

compute the values of the derivatives $f^{(k)}(0), k = 1, 2, 3, \ldots$.
Hint: Find a simple function q with the prescribed values and subtract it from f. What does repeated use of Rolle's theorem say about the derivatives of $f - q$ at 0?

Reminder PSet 3 to be released tomorrow, and due by the end of Oct 30 (on Canvas).